Ball on a Beam Control System: Evidence Report

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1 Background

1.1 Proportional-Integral-Derivative (PID) Control

A PID controller is a widely used feedback mechanism in industrial and academic control systems. It continuously calculates an error value $e(t)$ as the difference between a desired setpoint and a measured process variable, and applies a correction based on proportional (P), integral (I), and derivative (D) terms:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\,d\tau + K_d \frac{de(t)}{dt} \tag{1}$$

- **Proportional (K_p):** Reduces the steady-state error and speeds up the initial response, though excessive gain leads to oscillation or instability.
- **Integral (K_i):** Eliminates steady-state error by accumulating past errors over time, though it can introduce phase lag and overshoot.
- **Derivative (K_d):** Reacts to the rate of change of the error, adding damping that suppresses overshoot and improves stability.

1.2 Pole Placement Method

Pole placement is a model-based design method in which the controller gains are chosen so that the closed-loop poles sit at pre-determined locations in the complex plane. Matching the closed-loop characteristic equation to a desired polynomial, defined by a target natural frequency ω_n and damping ratio ζ, gives direct control over transient metrics such as settling time and percentage overshoot.

2 System Discussion and Modeling

2.1 System Overview

The ball-on-beam system consists of a ball free to roll along a straight track of length L, driven by an electric motor that tilts the beam by an angle θ. The system is inherently unstable and has integrating dynamics in position, so open-loop control is impractical.

2.2 Modeling Assumptions

The following assumptions simplify the model and allow the linear design:

- **Rigid Beam:** The beam is assumed perfectly rigid, so bending and elastic deflections are neglected.
- **Point Mass Ball:** The ball is treated as a point mass with rolling inertia, ignoring slip and friction losses.
- **Small Angle Approximation:** For the linear model, $\sin(\theta) \approx \theta$.

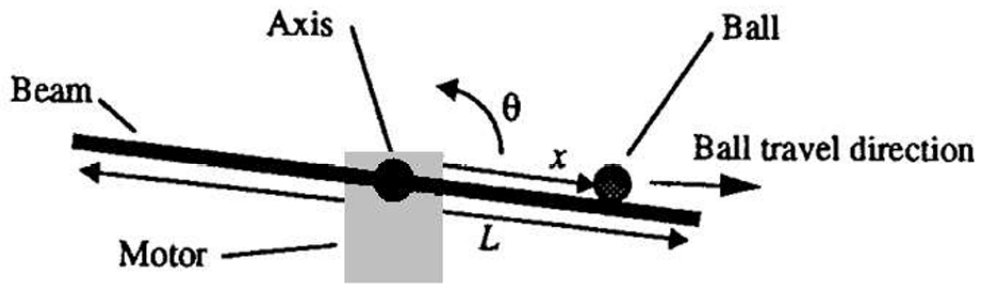


Figure 1: A figure showing the ball and beam system.

- **Constant Inertia:** The beam inertia J_0 is treated as constant in the linear domain.

The nonlinear simulation model is not used for the results in this report. The linear model is the basis of all reported performance figures.

2.3 Control Architecture and Block Diagram

A cascaded two-loop structure is used: an inner loop for fast angle tracking (θ) and an outer loop for ball position tracking (x).

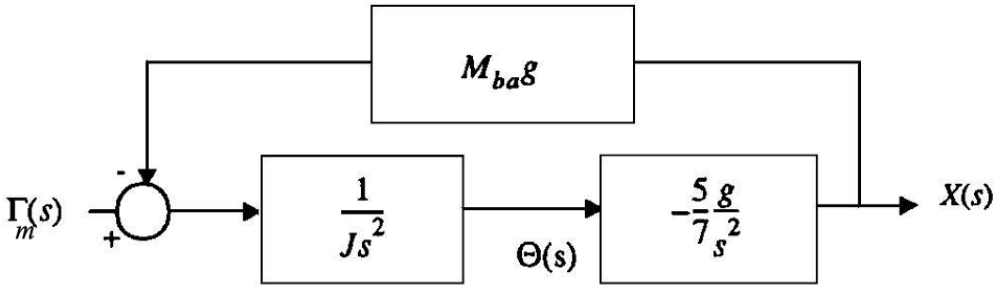


Figure 2: A block diagram for the ball and beam system.

2.4 Controller Design Overview

Designing a controller for an inherently unstable system with integrating dynamics like the ball on a beam requires careful handling of coupled behaviour. Rather than using a single multi-variable controller, the design is decomposed into a hierarchy. This separates the fast beam-angle dynamics from the slower ball-position dynamics.

2.5 Two-Loop Cascaded Architecture

- **Inner Loop (Angle Control):** Commands the motor torque to drive the beam angle θ to a desired angle θ_d. It operates at high bandwidth so the beam follows its command quickly.
- **Outer Loop (Position Control):** Computes the desired beam angle θ_d from the error between the ball's position x and the setpoint. It operates on a slower time scale.

Time-scale separation is required: the inner loop must be significantly faster than the outer loop so that the beam reaches its commanded angle before the ball position changes noticeably.

2.6 Sub-System Transfer Functions

The linearized equations of motion give two interconnected subsystems:

1. **Inner Angle Plant ($G_{inner}(s)$):** Relates the motor torque to the beam angle as a double integrator:

$$G_{inner}(s) = \frac{1}{J_0 s^2} \tag{2}$$

   where J_0 is the beam's rotational inertia. This is controlled by a PD controller $C_2(s) = K_{2d}s + K_{2p}$.

2. **Outer Position Plant ($G_{outer}(s)$):** Relates the beam angle to the ball's acceleration, $\ddot{x} = -c\,\theta$ with $c = \frac{5g}{7}$:

$$G_{outer}(s) = \frac{-c}{s^2} \tag{3}$$

   This is controlled by a PID controller $C_1(s) = K_{1d}s + K_{1p} + \frac{K_{1i}}{s}$.

2.7 Controller Equations

The outer loop determines the desired beam angle from the position error. With $h = \frac{5}{7g}$, this is

$$\theta_d(s) = \frac{5}{7g}\left(k_{1p} + \frac{k_{1i}}{s} + k_{1d}s\right)(X(s) - X_d(s)) \tag{4}$$

The inner loop computes the motor torque needed to tilt the beam while compensating for the ball's moment:

$$\Gamma_m(s) = M_{ball}\, g\, X(s) - J_0\left(k_{2p} + \frac{k_{2i}}{s} + k_{2d}s\right)(\Theta(s) - \Theta_d(s)) \tag{5}$$

where M_{ball} is the ball mass, J_0 is the beam inertia, and $k_{1,2}$ are the controller gains.

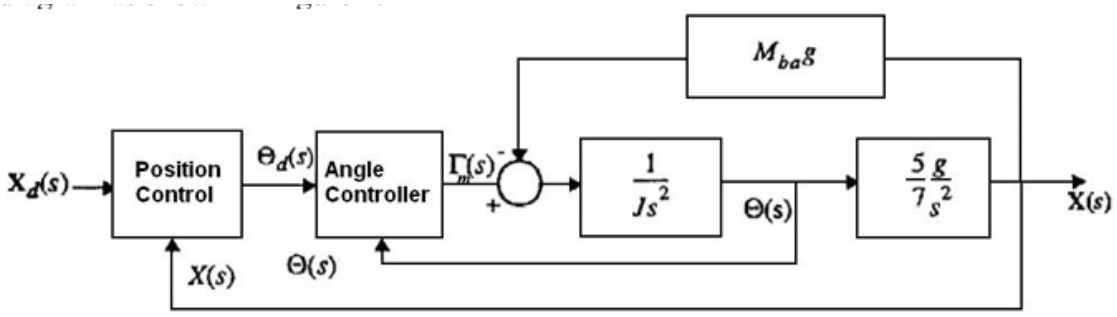


Figure 3: Complete system block diagram featuring the cascaded inner and outer loops.

2.8 Overall Characteristic Equation

Combining the outer PID controller $C_1(s)$, the inner controller $C_2(s)$, and the plant coefficients ($q = c \cdot h$), the closed-loop characteristic equation is

$$s^4 + s^2 C_2(s) + q\, C_1(s)\, C_2(s) = 0 \tag{6}$$

Expanding this gives the sixth-order polynomial used for pole placement:

$$s^6 + Ds^5 + (A + qdD)s^4 + (B + q(dA + aD))s^3 + q(dB + aA + bD)s^2 + q(aB + bA)s + qbB = 0 \tag{7}$$

where a, b, d are the outer PID gains (K_{1p}, K_{1i}, K_{1d}) and A, B, D are the inner gains (K_{2p}, K_{2i}, K_{2d}).

2.9 Setpoint Prefilter

With the PID gains placed for the target poles, the outer loop alone has a closed-loop numerator with zeros near the dominant poles. These zeros increase overshoot above the value predicted by the poles alone. To remove them, the setpoint is passed through a second-order prefilter before it reaches the outer loop:

$$F(s) = \frac{K_{1i}}{K_{1d}s^2 + K_{1p}s + K_{1i}} \tag{8}$$

The filter has unit DC gain, so the ball still settles at x_d. Its numerator cancels the outer-loop numerator, leaving the overall position response

$$\frac{X(s)}{X_d(s)} = \frac{K_{1i}}{s^3 + K_{1d}s^2 + K_{1p}s + K_{1i}} \tag{9}$$

whose poles are the designed outer poles at $-1.5 \pm 1.57j$ and -10. The prefilter is implemented as a state equation, $\ddot{r}_f = \left[K_{1i}(x_d - r_f) - K_{1p}\dot{r}_f\right]/K_{1d}$. The outer loop then tracks r_f rather than x_d.

3 Implementation, Simulation, and Results

3.1 MATLAB Implementation and Workflow

A standalone MATLAB script (`BallBeam_Cascade_Workflow.m`) implements the design and simulation. It performs the following steps:

- Defines the physical parameters of the ball-and-beam system.
- Computes controller gains for the inner angle loop and the outer position loop from the pole placement targets, with outer gains scaled by $1/q$.
- Simulates the linear closed-loop system over 6 seconds using `ode45`, with a 7-state model: ball position, velocity, beam angle, angular rate, integral of error, and the prefilter output and its rate.
- Computes overshoot, 5% settling time, peak beam angle, and peak motor torque from the simulated response.
- Runs the beam animation first, then displays the root loci and the position response figures.

3.2 Controller Gains and Performance Metrics

The outer loop poles were targeted at $-1.5 \pm 1.57j$ and -10, and the inner loop pole at -10 (double). The gains and measured results are:

Parameter	**Inner Loop (Angle)**	**Outer Loop (Position)**
Proportional Gain (K_p)	100.00	68.05
Integral Gain (K_i)	0.00	92.51
Derivative Gain (K_d)	20.00	25.48

Table 1: Designed controller gains.

3.3 Simulation Results and Root Loci

The figures below show the root loci for the cascaded loops and the closed-loop position response.

Metric	Simulated Value
Percentage Overshoot	4.84%
5% Settling Time (T_s)	1.440 s
Peak Beam Angle	7.454°
Peak Motor Torque	0.6743 N·m

Table 2: Measured linear closed-loop performance for $x_d = -0.25$ m from $x_0 = 0$ m.

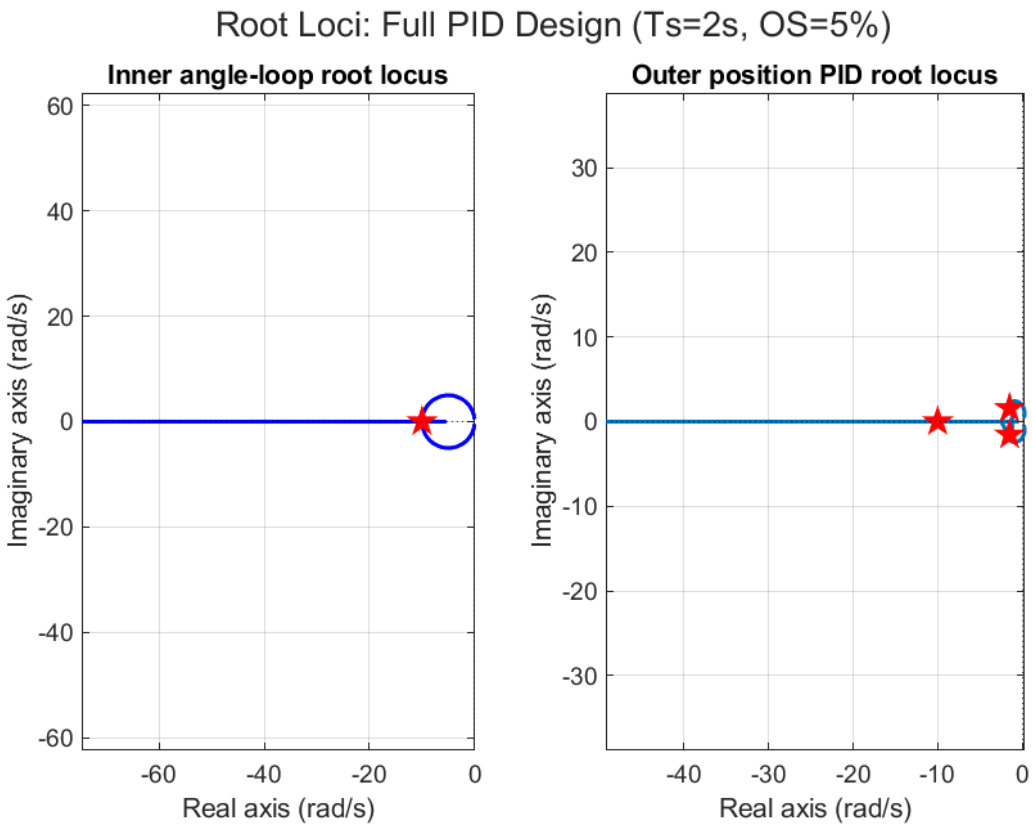


Figure 4: Root loci for the inner angle loop and outer position loop, showing the design pole targets.

3.4 Discussion of Results

- **Transient Performance:** The simulated overshoot of 4.84% is below the 5% target, and the 5% settling time of 1.44 s is below the 2 s target. Both specifications are met in simulation. The settling time is shorter than the 2 s design value, which is expected because the pole locations give a settling estimate of about $3/1.5 = 2$ s, and the prefilter and the ideal linear model change the response from that estimate. The simulation does not include actuator limits, sensor noise, or the nonlinear dynamics, so these figures should be treated as linear-model results.
- **Time-Scale Separation:** The inner loop pole at $s = -10$ is well to the left of the outer dominant poles at $-1.5 \pm 1.57j$, which supports the assumption that the loops can be designed separately.
- **Root Loci:** The root loci show the poles moving from their open-loop locations toward the design targets as the gain multiplier λ increases to 1. The outer loop branches approach the damping ratio $\zeta \approx 0.69$ associated with the dominant pair.
- **Control Effort:** The peak beam angle of 7.45° and peak motor torque of 0.674 N·m are reported from the linear simulation. Whether these are within the physical limits of the laboratory apparatus

has not been verified.

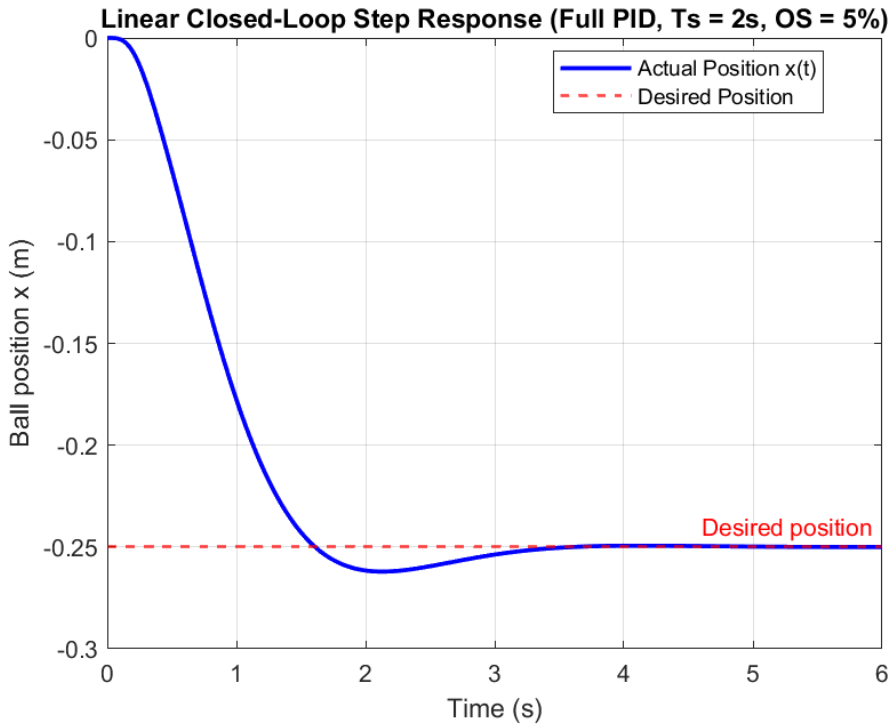


Figure 5: Linear closed-loop step response of the ball position $x(t)$ tracking $x_d = -0.25$ m.